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Problem Statement

Question : Find the equation of the line through $(-2,3)$ with slope -4 .

Equation of Line in Slope form

Solution : Given point is

$$\mathbf{h} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}, \text{Slope} = m = -4 \quad (3.1)$$

The equation of the line is given by

$$y = mx + c \quad (3.2)$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ mx + c \end{pmatrix} = \begin{pmatrix} 0 \\ c \end{pmatrix} + x \begin{pmatrix} 1 \\ m \end{pmatrix} \quad (3.3)$$

$$\mathbf{x} = \mathbf{h} + k\mathbf{m}, \mathbf{m} = \begin{pmatrix} 1 \\ m \end{pmatrix} \quad (3.4)$$

So

$$\mathbf{n}^T \mathbf{x} = \mathbf{n}^T \mathbf{h} = c \quad (3.5)$$

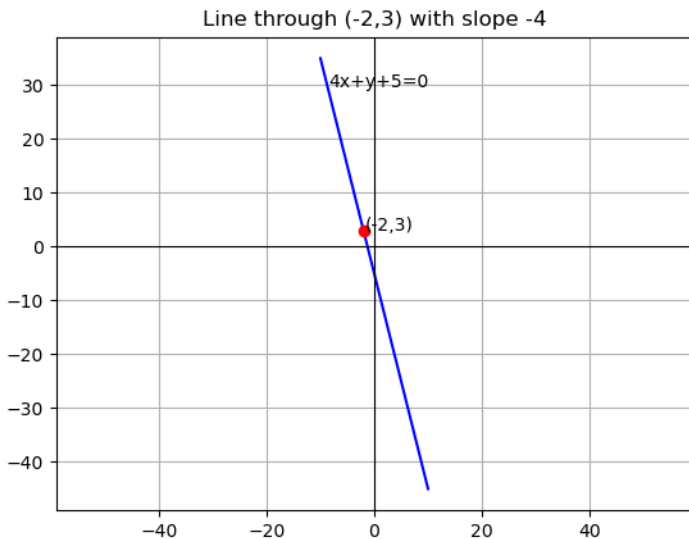
where $\mathbf{h}$ is any point on the line and $\mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix}$

$$c = \mathbf{n}^\top \mathbf{h} = (4 \ 1) \begin{pmatrix} -2 \\ 3 \end{pmatrix} = -5 \quad (3.6)$$

so equation of line is

$$y = -4x - 5 \implies \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ -5 \end{pmatrix} + x \begin{pmatrix} 1 \\ -4 \end{pmatrix} \quad (3.7)$$

Plots



C Code

```
#include <stdio.h>
// Function to print line equation given point (x0,y0) and slope
void line_equation(double x0, double y0, double slope) {
    // direction vector from slope
    double dx = 1.0;
    double dy = slope;
    // normal vector = (dy, -dx)
    double a = dy;
    double b = -dx;
    double c = -(a * x0 + b * y0);

    printf(" Point-on-line:-(%.2f,%.2f)\n", x0, y0);
    printf(" Slope:-%.2f\n", slope);
    printf(" Cartesian-form:-%.2fx-+-%.2fy-+-%.2f=-0\n", a, b, c);
    printf(" Vector-form:-r=-(%.2f,%.2f)-+t(%.2f,%.2f)\n",
           x0, y0, dx, dy);
}
```

```
// Function exposed for Python (shared object)
const char* get_line_equation() {
    static char result[200];
    double x0 = -2, y0 = 3, slope = -4;
    double dx = 1.0, dy = slope;
    double a = dy, b = -dx, c = -(a * x0 + b * y0);

    snprintf(result, sizeof(result),
        "Equation: %.2fx + %.2fy + %.2f = 0; Vector: r = (%.2f,
            %.2f) + t(%%.2f, %%.2f)",
        a, b, c, x0, y0, dx, dy);
    return result;}

int main() {
    // Example: line through (-2,3) with slope -4
    line_equation(-2, 3, -4);
    return 0;
}
```


Python : call_c.py

```
import ctypes
import os

# Part 1: Call the C shared object
# Path to shared object (must be in same directory)
lib_path = os.path.abspath("./libline.so")

# Load the shared library
lib = ctypes.CDLL(lib_path)

# Tell Python return type of the function
lib.get_line_equation.restype = ctypes.c_char_p
# Call the function
c_result = lib.get_line_equation().decode("utf-8")
print("=== Result from C shared library ===")
print(c_result)
```

```
print()
```

```
# Part 2: Direct computation in Python
```

```
def line_equation(x0, y0, slope):
```

```
    dx, dy = 1.0, slope
```

```
    a, b = dy, -dx
```

```
    c = -(a * x0 + b * y0)
```

```
    print("=== Direct Python Computation ===")
```

```
    print(f'Point on line: ({x0}, {y0})')
```

```
    print(f'Slope: {slope}')
```

```
    print(f'Cartesian form: {a:.2f}x + {b:.2f}y + {c:.2f} = 0')
```

```
    print(f'Vector form:  $r = ({x0}, {y0}) + t({dx}, {dy})$ ')
```

```
    print()
```

```
line_equation(-2, 3, -4)
```

```
# Part 3: Row reduction (manual)
```

```
def row_reduction():
```

```

print("=== Row Reduction Verification ===")
# Equations:  $a - 4b = 0$ ,  $-2a + 3b + c = 0$ 
print("System of equations:")
print("1)  $a - 4b = 0$ ")
print("2)  $-2a + 3b + c = 0$ ")

# From (1):  $a = 4b$ 
# Substitute into (2):  $-8b + 3b + c = 0 \Rightarrow -5b + c = 0 \Rightarrow c = 5b$ 
a = 4
b = 1
c = 5
print(f"Solution (up to scale): (a, b, c) = ({a}, {b}, {c})")
print(f"Equation: {a}x + {b}y + {c} = 0")
print()

row_reduction()

```

Python Code for Plotting

```
import numpy as np
import matplotlib.pyplot as plt
x=np.linspace(-10, 10, 100)
y=(-4*x)-5

plt.plot(x, y, '-b')
plt.plot(-2, 3, 'ro')
plt.text(-8.6, 29.6, "4x+y+5=0", fontsize=10, color="black")
plt.text(-1.8, 2.9, "(-2,3)", fontsize=10, color="black")
plt.axhline(0, color="black", linewidth=0.8)
plt.axvline(0, color="black", linewidth=0.8)
plt.title("Line through (-2,3) with slope -4")
plt.grid(True)
plt.axis("equal")
plt.savefig("../figs/fig6.png")
plt.show()
```